INCH-POUND

MIL-T-48621A (AR)
9 August 1991
SUPERSEDING
MIL-T-48621 (AR)
21 July 1980

MILITARY SPECIFICATION

TESTER, NOISE, X-BAND, AN/PSM-44: 11822700

This specification is approved for use by the U. S. Armament, Munitions and Chemical Command, and is available or use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. The Tester, Noise, x-Band, hereinafter referred to as the "Noise Tester, consists of microwave and electronic components necessary to provide a pass/fail indication of the FM Sideband Noise Power over the range of 600 Hz to 2 MHz from the carrier of a CW RF source operating anywhere in the frequency range from 9140 to 9250 Megahertz and having a power output of +10 dBm to +20 dBm maximum at the input port (RF IN) of the Noise Tester. The Noise Tester Is a Type I, Class 3, Style E, Color W, configuration as referenced in MIL-T-28800. This specification establishes the performance and test requirements for the Noise Tester, 13822700 (see 6.1). .

2* APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this . specification to -the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1290

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Material, Manufacture and Inspection, General Specification for
MIL-T-28800	Test Equipment for Use With Electrical and Electronic Equipment, General Specification for
MIL-I-45607 -	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-S-48431	Solid State X-Band Source: 10548182

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electro-magnetic Interference
MIL-STD-810 -	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 -	Standard Requirements for Soldering Electrical and Electronic Assemblies
MS3106	connector, Plus, Electric, Straight

(unless" otherwise indicated, copies of" federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Building 413, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS (See 6.4)

u. s. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548102 - Solid State X-Band Source
10549175 - Oscillator Module
10556651 - Power Source, Microwave
11822700 - Tester, Noise, X-Band, AN/PSM-44
11822701 - RF Modulator/MIA
11822702 - RF Detector/MIA
11822703 - Reference Oscillator
11822704 - Low Noise Amplifier
11822705 - Post Amp and RMS Detector
11822706 - LP Filter - 2 MHz
11822707 - Terminal Board

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

1182270$ - HP Filter - 750 KHz
11822709 - HP Filter - 2500 Hz
11822710 - Self Test Circuit
11822711 - HP Filter - 600 Hz
12822712 - Dc Filter/Regulators
11822713 - LP Filter- 750 KHz
11822714 - Video Amplifier
11822716 - Oper/Test Switch Assy
11822734 - Chassis assembly
11822737 - Dc Input Filter
11822738 - LP Filter - 2500 Hz
11822740 - Interconnection Diagram
11822752 - Cable W204
11822753 - Cable W201, W202, W203

SPECIAL PACKAGING INSTRUCTION

PS11822700 - Packaging of X-Band Noise Tester: 11822700

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedance. In the event of conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 11822700, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the noise tester shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.2.1 Subassemblies. The Noise Tester consists of seventeen major subassemblies which, when assembled into the Panel Assembly and Chassis, constitute the complete unit. Each subassembly is a replaceable part, and easily installed once the Noise Tester has been removed from its case. Troubleshooting and Self Test operations will isolate a fault to this subassembly level, permitting substitution/repair to be made at the Direct Support level of maintenance. The noise tester shall meet the requirements of the following drawings prior to find acceptance:

Functional Group Code	Subassembly	Drawing Number
1A1	chassis Assembly	11822734
lAIAl	RF Modulator/MIA	ll822701
IAIA2	RF Detector/MIA	11822702
1A1A3	Low Noise Amplifier	11822704
1A1A4	Post Amp & RMS Detector	11822705
lAlA5	LP Filter - 2 MHz	11822706
lAlA6	HP Fi1ter - 750 KHz	11822708
lAlA7	HP Filter - 2500 Hz	11822709
lAlA8	HP Filter - 600 Hz	l1822711
1A1A9	LP Filter - 750 KHz	11822713
lAIAlO	LP Filter - 2500 Hz	11822738
lAIAll	Video Amplifier	11822714
lA1A12	Dc Filter/Regulators	11822712
lA1Al3	Dc Input Filter	11822737
lA1Al4	Self Test Circuit	11822710
IA1A15	Terminal Board	11822707
lAlA16	Reference Oscillator	11822703
1A1Al7	Oper/Test Switch Assy	11822716

The Interconnection Diagram, Drawing 11822740, shows the final configuration of the above subassemblies, as they function in the complete noise tester.

3.2.1.1 Chassis assembly (1A1). The front panel of the Noise Tester shall consist of an etched 1/8 inch thick anodized and painted aluminum plate, with appropriate mounting holes for the Meter, the Cavity, the MODE Switch, the OPER/TEST Switch, all input and output connectors, ON/OFF Switch, and Fuze. Handles are mounted to the panel front, on either side. All installation and dimensions and specifications shall conform to Drawing 11822733.

3.2.1.2 RF modulator/MIA (lAlAl). The RF Modulator/MIA is a microwave integrated assembly consisting of an array of isolators, couplers, samplers, and attenuators, Interconnected with strip-type transmission line, which supply the reference level necessary for unit calibration. All dimensions and specifications shall conform to Drawing 11822701.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.2.1.3 RF detector/MIA (1A1A2). The RF Detector/MIA includes the cavity, input and output waveguides, and a microwave integrated assembly, consisting of the noise detector diodes, the phase shifter, the balance network, and couplers, all connected with stripline. All dimensions and specifications shall conform to Drawing 11822702.

3.2.1.4 Low noise amplifier(lAlA3). The Low Noise Amplifier is a three-stage amplifier with approximately 50 dB gain, with low noise characteristics that operates over the range of 600 Hz to 2 MHZ. All dimensions and specifications shall conform to Drawing 11822704.

3.2.1.5 Post amplifier and RMS detector (lAlA4). The Post Amplifier is single stage video amplifier that drives [illegible] that drives the RMS Detector which converts the AC signal to a DC level and, through another stage of amplification, drives the meter. All dimensions and specifications shall conform to Drawing 11822705.

3.2.1.6 Low pass filter - 2MHz (lAlA5]. The Low Pass Filter - 2 MHz is a passive low pass filter with a cutoff of 2 MHz in series With 750 KHz High Pass Filter (lAlA6), thus forming the Band 3 bandpass filter. All dimensions and specifications shall conform to Drawing 11822706.

3.2.1.7 High Pass filter - 750 KHZ (lAlA6). The High Pass Filter - 750 KHz is a passive high pass filter with a cutoff of 750 KHz, which in combination with 1A1A5 forms the Band 3 bandpass filter. All dimensions and specifications shall conform to Drawing 11822700.

3.2.1.8 High Pass filter - 2500 Hz (1A1A7). The High Pass Filter - 2500 Hz is an active high pass filter with a cutoff of 2500 Hz, which in combination with the Low Pass Filter - 750 KHz (1A1A9) forms the Band 2 bandpass filter. All dimensions and specifications shall conform to Drawing 11822709.

3.2.1.9 High Pass filter - 600 Hz (1AlA8). The High Pass Filter - 600 Hz is an active high pass filter with a cutoff of 600 Hz, which in combination with the Low Pass Filter - 2500 Hz (1A1A10) forms the Band 1 bandpass filter. All dimensions and specifications shall conform to Drawing 11822711.

3.2.1.10 Low Pass filter - 750 KHz (1A1A9). The Low Pass Filter - 750 KHz is a passive low pass filter with a cutoff of 750 KHZ, which in combination with 1A1A7 forms the Band 2 bandpass filter. All dimensions and specifications shall conform to Drawing 11822713.

3.2.1.11 Low Pass filter - 2500 Hz (1A1A10). The Low Pass Filter - 2500 Hz is an active low pass filter with a cutoff of 2500 Hz, which in combination with lAlA8 forms the Band 1 bandpass filter. All dimensions and specifications shall conform to Drawing 11822738.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.2.1.12 Video amplifier (lAlAll). The Video Amplifier is a one stage low video amplifier which takes the output from the LOW Noise Amplifier (1A1A3) and feeds it to the Noise Out jack on the front panel for external observation and testing. All dimensions and specifications shall conform to Drawing 11822714.

3.2.1.13 DC filter/regulators (1A1A12). The DC Filter cleans the incoming 24 ± 6VDC power line of line r hash, extraneous noise, and low-frequency transients. The regulators drop the Incoming 24 ± 6VDC to a stable 12 volts DC. All dimensions and specifications shall conform to Drawing 11822712.

3.2.1.14 DC input filter (lAlA13). The DC Input Filter is front panel mounted, and cleans the Incoming 24 + 6VDC line *of noise* within the operating bandwidth of the Noise Tests. All dimensions and specifications shall conform Drawing 11822737.

3.2.1.15 Self test circuit (lAlA14). The Self Test Circuit provides a means by which the gain of the filters and amplifiers for each band can be quickly checked. The out of lAlA16, the Reference Oscillator, is applied directlyto the inputs of the three bandpass filters *through an* appropriate attenuator thereby bypassing the RF Modulator/MIA (1A1A1). and-the RF Detector/MIA (1A1A2). The Reference Oscillator level is preset so that a quick analysis of individual band gain can be made by observing the meter. All dimensions and specifications shall conform to Drawing 11822710.

3.2.1.16 Terminal board (lAlA15). The Terminal Board is a discrete interface between the control clrcuits and the operating functions of the Noise Tester. all dimensions and specifications shall conform to Drawing 11822707.

3.2.1.17 Reference oscillator (1AlA16). The Reference Oscillator is a selectable three-frequency oscillator, operating at 2 KHZ, 43 KHZ,, and 1 MHZ, which provides the reference signal by which the Noise *Tester Is* initially calibrated. It is also used in the Self Test Circuit to provide a ready Indication of amplifier gain. All dimensions and specifications shall conform to Drawing 11822703.

3.2.1.18 Oper/test switch assembly (lAlA17). The Oper/Test Switch Assembly is a twelve-position printed circuit master function switch that orders all control functions of the Noise Tester. Al1 dimensions and specifications shall conform to Drawing 11822716.

3.2.2 Government-furnished material.

10548182	Solid State X-Band Source
10549175	Oscillator Module (six single channel modules, channels 1 through 6)
10556651	Power Source, Microwave
11748124	Tester, Radar Systems AN/VPM-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.3 Environmental service condition.

3.3.1 Baseline data. Prior to any environmental tests of the Noise Tester, the following test shall be performed at Standard Ambient conditions: With equipment connected as shown in Figure 3, a qualified good Microwave Power Source (Table 111, Item 18) shall be noise loaded at the O MOD INPUT port with an External Noise Source (Item 26), so that with an Input signal to the Noise Tester of +10 dBm at any frequency between 9140 and 9250 MHz, the NOISE INDICATOR *meter* will read within the limits defined by the black band when measuring the Source on any of the three bands. The levels of in injected noise required to achieve "black-line" for each band (as read on the True RMS Voltmeter, Item 28) end the power level indicated on the input monitoring Power Meter (Items 6, 8) shall be recorded as Baseline Data.

3.3.2 Repeatability. Subsequent to any environmental testing of the Noise Tester, the following test shall be performed: With equipment connected as shown in Figure 3, the Baseline Data shall be noise loaded with the same level of External Noise as recorded in the Baseline Data for the band under test. The Variable Waveguide Attenuator (Item 16) shall be adjusted to obtain a black line center meter reading on the Noise Tester, when it is set up for noise measurement of the band under test. The resultant level indication on the Input monitoring Power Meter (Items 6, 8) shall be within ± 2 dB of the same level recorded in the Baseline Data.

3.3.3 Physical characteristics. The Noise Tester is housed in a Style E enclosure, as defined in MIL-T-28800. The enclosure has external dimensions of 12 In H x 16 In W x 15.5 in L, and has an internal volume of 1.7 cubic feet. The Noise Tester in its enclosure weighs 53 pounds.

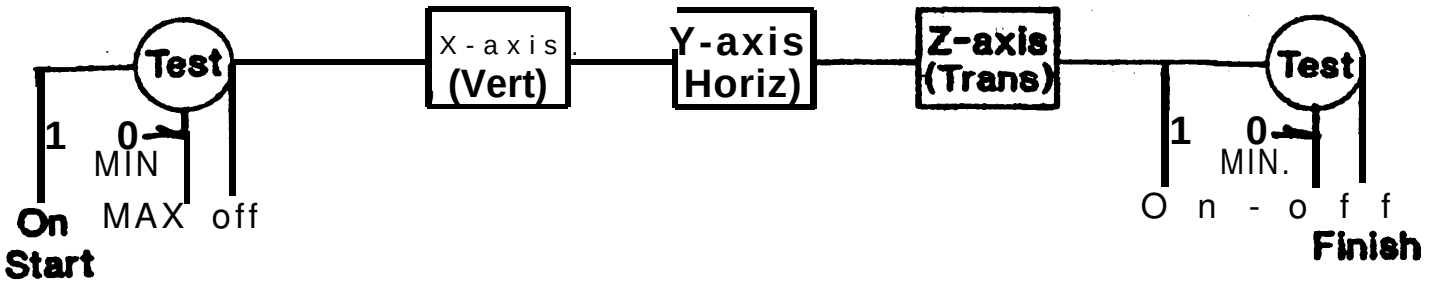
3.3.4 Reliability. The Noise Tester shall be capable of meeting the *requirements* for Type I reliability as defined in MIL-T-28800, i.e., a minimum MTBF (Mean Time Between Failure) of 1000 hours for continuous and intermittent operation, for the environmental conditions *specified* herein 3.3.6 - 3.3.1.3.

3.3.5 Maintainability. The Noise Tester shall consist of seventeen non-repairable major subassemblies, all replaceable at the Direct Support level of maintenance. As such, the Noise Tester shall meet the requirements for Type I maintainability as defined in MIL-T-28800 .

3.3.6 Environmental. Pretest requirements as referenced in MIL-STD-810, paragraph 3.2, shall be confined to the retirements of paragraph 3.3.2 of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

3.3.7 Vibration. The Noise Tester shall be capable of withstanding vibration test requirements as defined in MIL-STD-810. Method 514, Procedure I, as modified in MIL-T-28800, paragraph 4.5.5.3.1, in each of three mutually perpendicular axes. Subsequent to the vibration test, the Noise Tester shall not have been damaged or adversely affected and shall meet all requirements of 3.,3.2 (see chart below).



e :

TEST TO PARA. 3.3.2

Shock (3.3.8) and/or Vibration (3.3.7)

TEST SEQUENCE

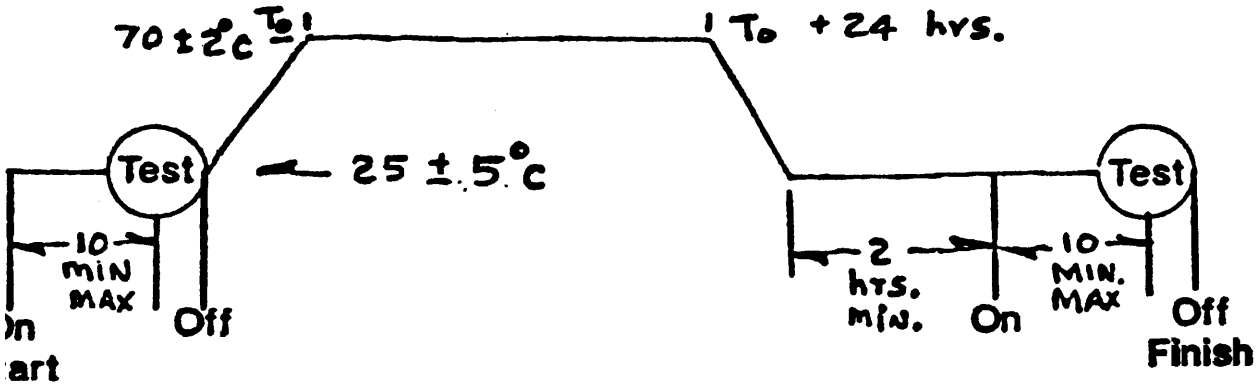
3.3.8 Shock. The Noise Tester shall be capable of withstanding the shock test requirements of MIL-STD-810, Method 516, Procedure 1, without being damaged or adversely affected. The Noise Tester shall receive six shocks in each direction of three mutually perpendicular axes. The shock pulse shall be 15g, 11 millisecond, 1/2 sine wave. subsequent to the shock test, the Noise Tester shall not have been damaged or adversely affected and shall meet all requirements of 3.3.2 (see chart in 3.3.7).

3.3.9 Altitude (operating and non-operating) The Noise Tester shall be capable of withstanding the altitude test requirements defined in MIL-STD-810, Method 500, Procedure I, except that the simulated altitudes shall be as specified in MIL-T-28800, Table XIII, for Class 4 equipment. Subsequent to the altitude tests, the Noise Tester shall not have been damaged or adversely affected and shall meet all requirements of 3.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.3.10 Temperature.

3.3.10.1 Temperature. (non-operating). The Noise Tester shall be capable of withstanding the temperature test requirements as defined in MIL-STD-810, Method 501, Procedure 1, without being damaged or adversely affected. The Noise Tester shall be temperature stabilized at +70 ± 2 degrees C and -62 ± degrees C for a period of 24 hours at each temperature extreme. After each of the above temperature tests, the Noise Tester shall be temperature stabilized at 25 ± 5 degrees C for at least two hours. Subsequent to the temperature stabilization period, the Noise Tester shall not have been damaged or adversely affected and shall meet all requirements of 3.3.2 (see charts below).



NON-OPERATING HIGH TEMPERATURE TEST SEQUENCE

Finish

To + 24 hrs.

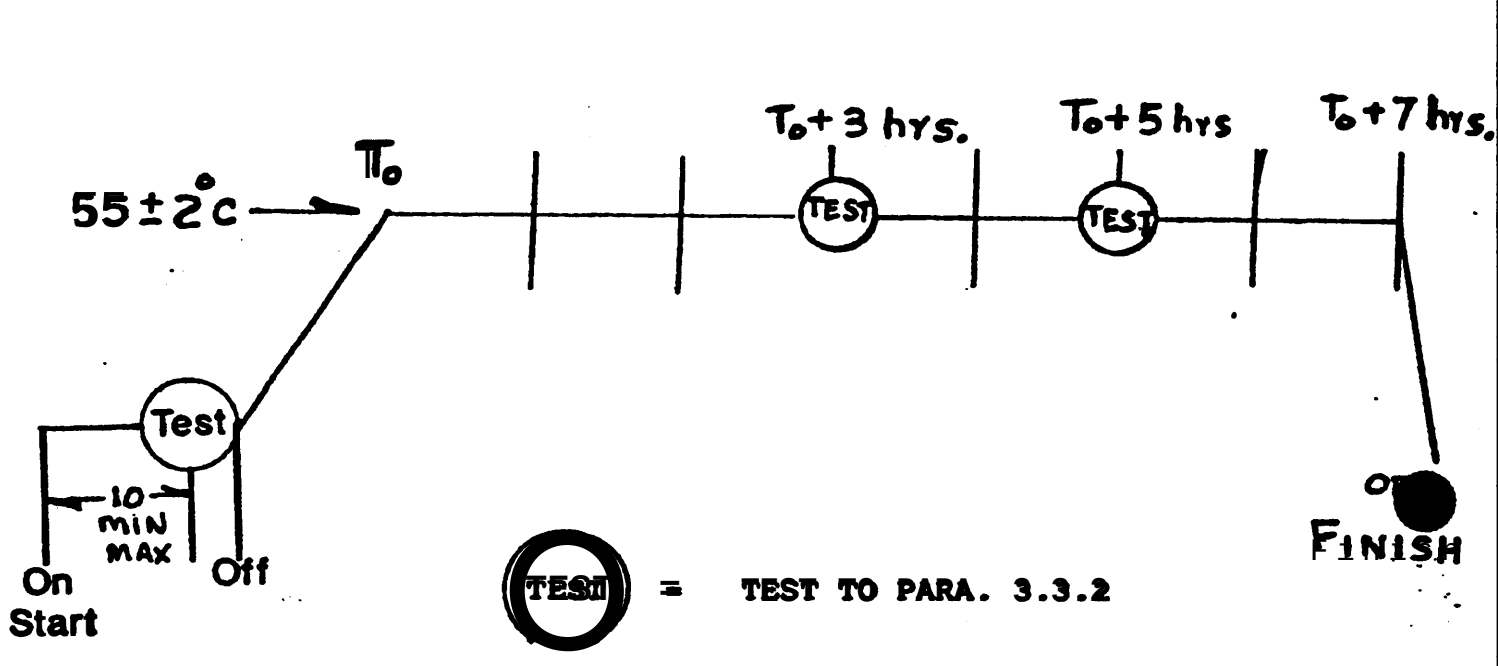
NON-OPERATING LOW TEMPERATURE TEST SEQUENCE

TEST = Test to para 3.3.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.3.10.2 Temperature (operating).

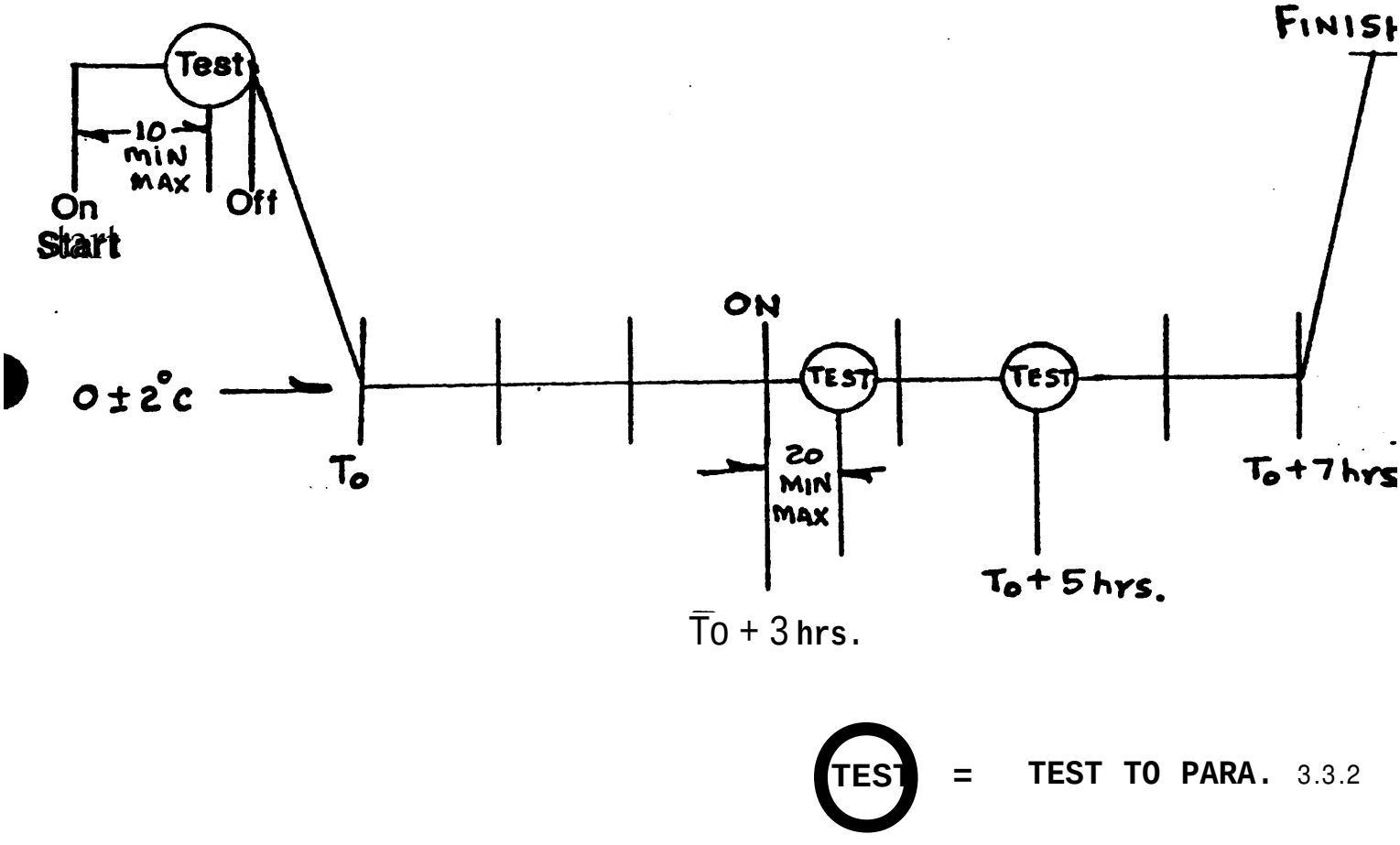
3.3.10.2.1 High temperature. The Noise Tester shall be capable of withstanding the temperature test requirements as defined in MIL-STD-810, Method 501, Procedure I, without being damaged or adversely affected. The Noise Tester, after a minimum of three hours temperature stabilization at +55 ± 2 degrees C with nominal primary power applied (+24 ± 1VDC), shall be capable of continuous operation for four hours. During the four-hour operating Interval, the Noise Tester shall be tested in the first and third hours to meet all requirements of 3.3.2 (see chart below). I



HIGH OPERATING TEMPERATURE TEST SEQUENCE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.3.10.2.2 **Low temperature.** The Noise Tester shall be capable of withstanding the temperature test requirements as defined in MIL-STD-810, Method 502, Procedure I, without being damaged or adversely affected. The Noise Tester, after a minimum of three hours temperature stabilization at 0 ± 2 degrees C, without primary power applied, shall be capable of continuous operation for four hours. During the four-hour operating interval, the Noise Tester shall be tested 20 minutes after nominal primary power Is applied (+24 ± 1VDC), and during the third hour of the operating test interval to meet all requirements of 3.3.2 (see chart below).



LOW OPERATING TEMPERATURE TEST SEQUENCE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.3.11 Humidity (non-operating) The Noise Tester shall be capable of withstanding the humidity test requirements defined in MIL-STD-810, Method 507, Procedure IV, as modified in MIL-T-28800, paragraph 4.5.5.1.1.2, (a) through (g), without being damaged or adversely affected. Subsequent to the humidity test and removal of excess moisture, the Noise Tester shall not have been damaged or adversely affected and shall meet all reguirements of 3.3.2.

3.3.12 Fungus resistance. The Noise Tester shall be capable of withstanding the fungus test of MIL-STD-810, Method 508, *Procedure I* (28 days), as modified in MIL-T-28800, paragraph 3.9.7. Subsequent to the fungus test, the Noise Tester shall not have been damaged or adversely affected and shall meet all requirements of 3.3.2.

3.3013 Bench handling. The Noise Tester shall be capable *of* withstanding the bench handling test requirements as defined in MIL-STD-810, Method 516, Procedure V, without being damaged *or* adversely Subsequent to the bench handling test, the Noise Tester shall not have been damaged or adversely affected, and shall meet all requirements of 3.3.2.

3.3.14 EMI/RFI. The Noise Tester shall be capable of withstanding the EMI/RFI test requirements as defined in MIL-STD-461, CE03, CSO1, CS02, RE02, and RS03 except the upper limit of RS03 shall be .1 V/M, because the system Is deployed in the AN/TSM-115 shelter which is shielded inclosure.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the noise tester.

3.5 Performance.

3.5.1 Warm-up. An initial warm-up period of ten to fifteen minutes shall occur prior to the commencement of any performance testing unless otherwise specified.

3.5.2 Primary input power. The noise tester shall be capable of meeting all requirements of the specification with +24 ± 6 Vdc primary input power (maximum input current 0.6 amp).

a. The following requirements shall be met with the power supply cable W204 connected between a +24 ± 6 Vdc at 0.6A power source and the front panel voltage input connector marked +24 Vdc input.

b. With MODE switch set to INTL TEST and the OPER/TEST switch set to either +12 VA or +12 VB, the meter (Ml) shall remain at center position (i.e., within the limits defined by the black band), over an input voltage range of +18 to +30 Vdc at 0.6 ampere, as measured at the +24 Vdc test points on the front panel.

Pilot lamp DS1 shall remain illuminated at a constant intensity over the full range of input operating voltage (+24 ± 6 Vdc at 0.6A].

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

3.5.3 RF IN.

a. The operating input power range shall be 10 milliwatts (+10 DBM) minimum, and 100 milliwatts (+20 DBM) maximum.

b. The maximum allowable non-operating input power shall be 200 milliwatts (+23 DBM).

The above operating limits shall be met over the frequency range of 9145 to 9250 MHz.

3.5.4 RF OUT.

For a signal input of 10 milliwatts (+10 DBM), with the RF LEVEL control set to its fully clockwise position, the signal level at the RF OUT connector shall be -9 + 2 DBM. With the RF LEVEL control set fully counterclockwise, the signal level at the RF OUT connector shall be at least 12 DB down from the reading taken with the RF LEVEL control fully clockwise.

b. With the MODE switch in the INTL or EXT TEST mode, and the OPER/TEST switch in OPERATE BAND 1, BAND 2, or BAND 3, the level of FM modulation on the signal by the reference oscillator shall be adjustable over the range of 27 to 33 DB below the carrier, as determined by the setting of the RF LEVEL control.

3.5.5 NOISE OUT.

With the MODE switch in the EXT TEST or OPER position, the NOISE OUT Port shall provide a demodulated signal from the FM demodulator suitable for external spectral analysis, of other measurements.

b. With the MODE switch in the EXT TEST position and the OPER/TEST switch set to +12 VB, the NOISE OUT port shall be capable of providing a 0.8 volt peak-to-peak minimum signal terminated into 50 ohms.

With the MODE switch in the EXT TEST or OPER mode, the NOISE OUT port shall provide a signal for the measurement of the internal reference attenuators, as dictated by the positions of the OPER/TEST switch:

	OPER/TEST switch position	
Band	Attenuator Out	Attenuator In
1	+12 VA	BD 1 TEST
2	+12 VB	BD 2 TEST
3	MTR OFF	BD 3 TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

d. The noise tester shall be capable of phase modulating the source being tested with the following reference frequencies:

Band	Modulation Frequency
1	2000 Hz
2	43 KHz
3	1.0 MHz

Each modulation frequency shall be within + 10 percent of each above specified frequency. Each signal shall be verifiable for frequency . at the NOISE OUT port, using an external frequency counter.

3.5.6 RF tuning controls. ,,,

The FREQ TUNE control shall be capable of tuning over the range of 9145 to 9250 MHz, as indicated by a peaking effect on the noise indicator meter (Ml) with the OPER/TEST switch in the CAV TUNE position, and the MODE switch in the INT or EXT TEST position.

b. The BAL ADJ control shall be capable of matching the output of the two internal detector diodes, as indicated by a black cantor line reading on the meter (Ml), with the OPER/TEST switch in the BAL position, and the MODE switch in the INT or EXT TEST position.

c. The PHASE ADJ control-shall be capable of adjusting the FM demodulator to phase quadrature as Indicated by a black center line reading on the meter (Ml), with the OPER/TEST switch in the PHASE SHIFT position, and the MODE switch in the INT or EXT TEST position .

3.5.7 System gain self test. The noise tester shall have as a self test function a means of verifying the gain integrity of each band, as indicated by a blue band indication on meter Ml, with the following control settings: MODE switch set to INTL TEST, the OPER/TEST switch in the BAND 1, BAND 2, or RAND 3 test positions, the OSC TEST pushbutton depressed, the REP LEVEL control adjusted for black center line, and the OSC TEST pushbutton then released.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

3.5.8 Integrated noise measurement.

The noise tester shall be capable of verifying the integrated noise as derived from MIL-S-48431, "Solid State X-Band Source" Figure 3. The noise power shall be integrated in three bands as follows:

BAND 1: 600 to 2500 Hz
BAND 2: 2500 to 75000 Hz
BAND 3: 750 to 2000 KHz

Tolerances for frequency limits of each band are per the following drawings:

BAND 1: 11822711, 11822738
BAND 2: 11822709. 11822713
BAND 3: 11822708, 11822706

b. The black center line of the noise Indicator meter shall represent the total Integrated noise power for each band (as derived from MIL-S-48431) as follows:

BAND 1: -54.3 DB below the carrier
BAND 2: -57.3 DB below the carrier
BAND 3: -56.8 DB below the carrier

The tolerance for the Integrated noise power measurement for each band shall be ± 2 DB.

The noise tester AN/PSM-44 when positioned 12 inches from speaker LS305 and when tested in accordance with 4.7.8.11 and 4.7.8.12 In its normal configuration, shall display a green area meter reading when pseudo-random noise of 6.0 volts RMS at Band 1 is injected into speaker LS305.

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections sat forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirmenents.

4.1.1 ResPonsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall Inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or suplies submitted to the Government for acceptance comp ly with all requirements of the contract. Sampling in qua lity conformance does not authorize submission of known defective material, either Indicated or actual, nor does It commit the Government to acceptanc~ of defective material.

4.1.2 General Provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shallbe made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of Inspections. The Inspection requirements Specified herein are Classified as follows :

a. First article inspection (see.4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 oII conditions. Unless otherwise specified, all examinations and conducted at the following standard ambient conditions:

a.	Temperature	+60 deg to +90 deg F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:34Z
Check the source to verify that this is the currentversion before use.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) noise testers for inspection in accordance with 4.3.2.

4.3.2 Inspections to be Performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified In Table I (functional tests) and Table II (Special sampling) herein and to any or all requirements of the applicable drawings.

If any assembly, component or test specimen fails to comply any of the applicable requirements, the first article sampl e shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of prod uct from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities resented for inspection. Homogeneity shall be considered to exist provided the Inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, In accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot .

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests (100 percent inspecticg.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	W A
Maj or		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7*3
Minor:	NONE	N/A

4.5 *special* sampling. One noise tester out of the first *so* produced and one out of each 100 produced thereafter shall be selected at random.. The sample shall have met the requirements and tests of Table I and shall then meet the tests In Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	NOne	N/A
Major		
103. Vibration	3.3.7	4.7.1.1
104. Shock	3.3.8	4.7.1.2
105. Altitude	3*3.9	4.7.1.3
106 . Temperature	3.3.10	4.7.1.4
107. Humidity	3.3.11	4.7.1.5
108. fungus	3.3.12	4.7.1.6
109. Bench handling	3.3.13	4.7.1.7
110. EMI/RFI	3.3014	4.7.1.8
Minor:	None	N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4*5*1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 InspeCtion equipment .

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-fuX test ent, supply and maintenance of test equipment shall be in accordance with the applicable requirements MIL-I-45607.

4.6.2 contractor-furnished Inspection equipment

4.6.2.1 *Government* design Unless otherwise specified in the contract, all inspection equipment identified by drawing number In specification or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval In accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the 'Methods of ispection" specified in 4.7 of this specification and with the component inspection procedures specified in 'Examination" and "Task Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4~6.2.1). since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used In the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive reflection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III (Test Equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE 111. Test equipment list.

ITEM NUMBER .	DESCRIPTION
1. Power supply	TYPE Laboratory supply. Minimum requirements : 20 to 30 Vdc. Load current capability: 1.5A. Ripple and noise: l MV rms to 5 mV peak-to-peak maximum from dc to 20 MHz
2. Power supply	Type Laboratory supply. Minimum requirements: 18 to 30 Vdc. Load current capability: 0.5A. Ripple and *noise:* 1 MV RMS to 5 mv peak-to-peak maximum from dc to 20 MHz. .
3* Ammeter, Dc	0-1.5A Accuracy: 2 percent
4-5. Digital Voltmeters, 3-1/2 digits minimum	Range: 0-100 Vdc minimum. Minimum accuracy: 0.1 percent *of* reading, +0.1 percent of scale.
6-7. Thermistor Mounts (compatible With Items 8 and 9)	Frequency *range:* 9145 to 9250 MHz minimum. Max- average Input RF *power:* 10 MW minimum.
8-9. Power Meters (compatible with items 6 and 7)	Frequency range: 9145 to 9250 MHz minimum. Power range: 1 mw full scale. Accuracy:+ 1 percent of full scale. Type: Thermistor sensing element.
10. Directional Coupler (10DB)	Bandwidth: 9145 to 9250 MHz minimum. Peak power rating: 2000W minimum. Coupling: 10 ± 0.4 DB. Mainline VSWR: 1.0:1.1 maximum. Secondary line VSWR: 1.0:1.2 maximum.
11. Precision Coaxial Attenuator - 10DB	Frequency range: 9145 to 9250 MHZ minimum. Attenuation: 10 ± 0.3 DB. Average power: 2 SCW minimum. MaXi- VSWR: 1.0:1.15.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE XIX. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
12.	Wava guide-to-OSM/SMA Adapter	Frquenq range: 9145 to 9250 MHz minimum. Maximum VSWR: 1.0:1.25.
13-14.	Waveguide-to-type N Adapters	Frequency range: 9145 to 9250 MHz minimum. Maximum VSWR: 1.0:1.25.
15•	Termination	Frequency range: 9145 to 9250 MHz minimum. VSWR: 1.0:1.05 maximum. Average power: lW minimum.
16.	Variable Waveguide Attenuator	Frequency range: 9145 to 9250 MHz minimum. Attenuation: O to 60 DB. Average power: 10W minimum. Peak power: 2KW minimum. Maximum VSWR: 1.0:1.1s.
17.	Variable Attenuator	*Frequency range:* 300 Hz to 3.0 MHZ minimum. Attenuation: O to 40 DB. Average power lW minimum.
18.	Microwave Power *Source*	Drawing 10556651.
19•	Solid State X-Band source	Drawing 10548182.
20-25.	Oscillator Modules	Drawing 10549175.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list (continued).

ITEM NUMBER	DESCRIPTION
26. External Noise Source (pseudo-random)	RDL Model NO. TE200A261 Specification: Band 1 Output Adjustable from: 10 to 600 UV ± (Output impedance: 50 ohm ± 20 Response: 600 to 2500 Hz flat within ± 5 DB 2000 Hz output-et to 600 ± 60 uV/ Hz Output of band rejection: 5000 to 10,000 Hz *relative* output 50 DB Band 2 and 3 Outputs Adjustable from: 2to 25uV ± Output impedance: 50 ohm ± 20 Response: 2500 to 750,000 Hz flat within ± 5 DB 43,000 Hz set to 25 ± 3 UV ± Response: 750,000 to 2,000,000 Hz flat within ± S DB 1,000,000 Hz set to 25 + 3 V ±
27. Audio Oscillator	Frequency range: 300 Hz to 3.0 MHz minimum. Power output: 10 Vp-p maximum. Impedance: 50 ohms. Waveform: Sine wave. Sine distortion: Less than 1 percent.
28. True RMS Voltmeter	Frequency range: 30 Hz to 3.0 MHZ minimum. Input sensitivity: 1 mV to 10 v. Accuracy: 1 percent full scale, or 2 percent of reading. Input impedance: 2 M Ohm.
29. Frequency Selective Voltmeter	Frequency range: 300 Hz to 3.0 MHZ minimum. Input sensitivity: 3 Uv to 30 V minimum. Input impedance: Bridged. Selectivity: 250 ± 50 Hz at 3 DB down, 1000 ± 100 Hz at 60 DB down. Accuracy: ± 0.2 DB.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
30.	Base Band Spectrum Analyzer	Frequency range: 300 Hz to 50 KHz minimum. Amplitude accuracy (log): 5 Hz-SO KHZ: ± 0.5 DB. Display range: 80 DB minimum.
31.	RF Spectrum Analyzer	Frequency range: 9145 to 9250 MHZ minimum. Dispersion: 1 KHz/a to 1 MIWcm . Bandwidth: 100 Hz to 300 KHz. Amplitude accuracy (log): ± 0.25 DB.
32.	Microwave Amplifier	Frequency range: 7.0-12.4 GHZ. Input level (max): 100 mw. Gain: 30 DB minimum.
33.	Precision Coaxial Attenuator - 3 DB	Frequency range: 9145 to 9250 MHz minimum. Attenuation: 3 DB ± 0.3 DB . Average power: 2W minimum. Peak power; 2 KW minimum. Maximum VSWR : 1.0:1.15.
34•	Precision Coaxial *Attenuator* - 20 DB	Frequency range: 9145 to 92S0 MHz minimum. Attenuation: 20 DB ± 0.3 DB. Average power: 2W minimum. Peak power: 2 KW minimum. Maximum VSWR : 100:1.15.
35.	oscilloscope	Vertical display: Single trace, dc to 10 MHz (3 DB down minimum) at 5 mV/div, minimum. Horizontal sweep, calibrated time base 50 ns/division to 1 s/division, minimum.
36.	BNC Tee Fitting	UG 274/U
37*	Electronic Frequency Counter	Frequency range: 300 Hz to 3 MHz minimum. Accuracy: + 0.1 percent minimum. Input sensitivity: 100 mV minimum.
38.	Power Supply	WP = Laboratory supply. Min$_L$= requirements: 10 to 30 vdc, current capability: 6A. Ripple and noise: l mV rms to 5mV peak-to-peak maximum from dc to 20 MHz .
39.	Calibrated Divider Network	RDL Drawing: 440006

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
40•	Power Cable Connector Adapter	MS3106A16-10S
41•	Cable w201	Drawing: 11822753
42.	Cable w202	Drawing: 11822753
43.	cable W203	Drawing : 11822753
44•	Cable W204	Drawing : 118227S2
45.	Speaker LS305	MIL-L-19011/l
46.	Amplifier, Realistic MPA- 35	Specifications: Input impedance: 50 K Ohms Output power: 35 watts (8 ohm 5% THD, 20 to 7 KHz) Input sensitivity (for 35 watt rated power): 150 mV Frequency response: 150 Hz - 12 KHz (-3 DB down) Signal /noise ratio: 65 DB. Signal/hum ratio: 65 DB. Hum & noise level: 5 mV. Distortion: 1.5% (A x input, 2 watts output Into 8 ohm outputs : High : 70 volt line

4.7 Methods of Inspection.

4.7.1 Environmental. Environmental tests shall be conducted - In accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 1, Procedure I, of MIL-STD-810 as modified by 3.3.7 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2. I apply. The noise tester shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the noise tester shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock Impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ±10% from 20 ± 2,000 hertz.

After the application all the shock impulses the noise tester shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.8.

4.7.1.3 Altitude (operating and non-operating) Thl& test shall be performed as specified in Procedures I, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.9.

4.7.1.4 Temperature.

4.7.1.4.1 Highoperating and non-operating) This test shall be pezformed aa specified in Procedures I and IX, Method 501 of MIL-STD-810 with constant temperature extremes as specifieed h 3.3.10. The operational checkout shall be as specified In 4.7.2 and 4.7.3 for compliance with 3.3.10.

4.7.1.4.2 Low (operating and non-operating) This test shall be performed as specified in Procedures I and II, Method 502 Of MIL-STD-810 with constant temperature extremes as specoified in 3.3.1o. The opera%%onal checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.10.

4.7.1.5 Humidity (non-operating This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the Item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.11.

4.7.1.6 Fungus. The fungus test shall be performed in accordance with MIL-STD-810, Method 508, Procedure 1.

4.7.1.7 Bench handling. The bench handling test shall be performed in accordance with MIL-STD-810, Method 516, Procedure I.

4.7.1.8 EMI/RFI. The EMI/RFI shall be performed IAW MIL-STD-461 CE03, CSC)l, CS02, CS06, RE02 and RS03 except the upper limit of RS03 shall be 0.1 V/M.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Initial conditions.

a. Connect cable W204 as shown in Figure 1.

b. Apply +24 Vdc ± 1 percent at 0.6 ampere (unless otherwise specified) to the power input cable W204, pins A(+), B(RET) and C(GND). For bench use, use connecter adapter (item 40, Table III) with extension leads soldered to pin terminals.

c. Turn ON/OFF switch to ON position.

d. Allow an initial warm-up as defined in 3.4.2.

4.7.3.2 Input voltage.

a. Set noise tester controls as follows:

(1) OPER/TEST switch to +12 VA position.

(2) MODE switch in either INT or EXT TEST position.

b. Meter (Ml) shall remain at center position (within limits defined by black band).

c. Set OPER/TEST switch to +12 VB position.

d. Meter (Ml) shall remain at center position (within limits defined by black band).

e. Set the power supply (item 2, Table 111) to +18 Vdc ~ 1 percent, as monitored on the DVM (item 5, Table III).

f* Repeat Steps a. through f.

9* Set the power supply (Item 2, Table III) to +30 Vdc ± 1 percent, as monitored on the DVM (item 5, Table III).

h. Repeat Steps a. through f.

i. The power indicator light (DS1) shall exhibit no perceptible change in intensity over the full range of input operating voltage (+24 ± 6 Vdc) as tested above.

j. Set the power supply (item 2, Table III) to +24 ± 1 Vdc as monitored on the DVM (Item 5, Table III).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

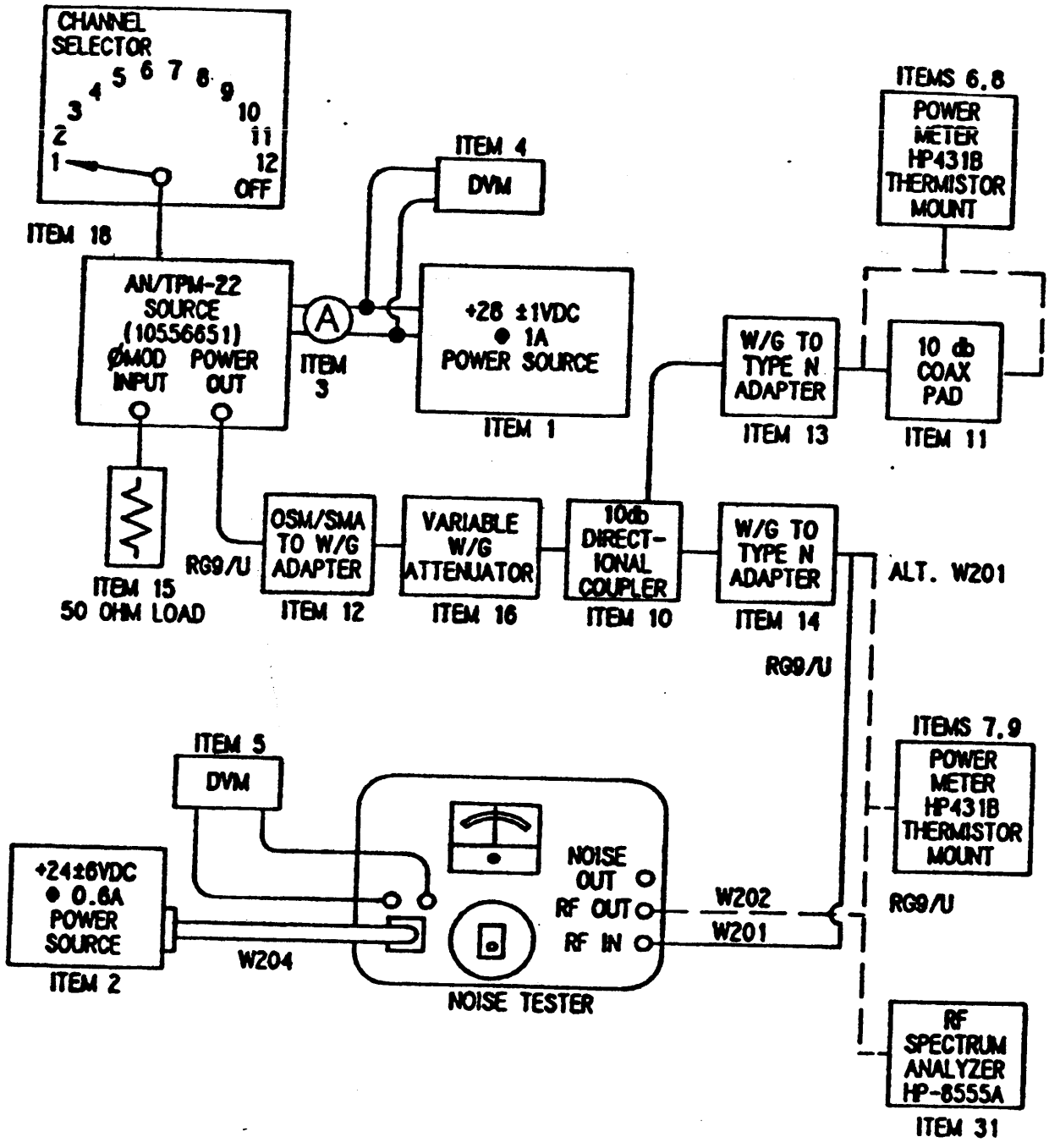


FIGURE 1. Initial test conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

4.7.3.3 RF IN.

a. Set up equipment as shown in Figure 1, with one power meter (Items 6 and 8, Table 111) connected to the W/G-to-Type N adapter (item 13, Table III) and the other power meter (items 7 and 9, Table III) connected to the waveguide-to-Type N adapter (item 14, Table 111) at the output of the directional coupler (item 10, Table III) through coaxial cable w201 (item 41, Table III).

b. Set the microwave power source (item 18, Table 111) channel selector to channel 1 (see Table IV, Frequency conversion charts). Apply +28 + 1 Vdc at 1A from the power supply (item 1, Table XII) as monitored by the DVM (item 4, Table III) and ameter (item 3, Table III).

c. Adjust the variable waveguide attenuator (item 16, Table III) for a reading of +10 DBM on the power meter (items 7 and 9, Table III) connected to cable W201 (Item 41, Table III).

d. Read the power meter (items 6 and 8, Table 111) connected to the W/G-to-Type N adapter (item 13, Table 111) and record this power level in column 2 of Table IV.

e. Repeat steps c. and d. for all twelve channels of the microwave power source (item 18, Table III).

f. Set the *microwave power* source (item 18, Table III) channel selector to Channel 1.

g. Disconnect the power meter (Items 7 and 9, Table 111) from cable W201 and connect W201 to RF IN port of the noise tester.

h. Using Table V (Output signals) as a guide, adjust the variable waveguide attenuator (Item 16, Table 111) for +10 DBM Input signal to the noise tester, as indicated on the power meter (items 6 and 8, Table III).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE IV. Frequency conversion charts.

RADAR : AN/VPS-2 (10548182~

OSCILLATOR MODULE CHANNEL NO.	OUTPUT FREQUENCY
CH1	9145 MHz
cH2	9153 MHz
CH 3	9161 MHZ
CH4	9169 i@iZ
CH5	9177 MHz
CH6	9185 MHZ

POWER SOURCE: AN/TPM-22 [10556651)

SWITCH POSITION	OUTPUT FREQUENCY
Ch 1	9145 MHz
Ch 2	9153 MHz
ch3	9161 MHz
CH4	9169 MHz
CH5	9177 MHz
CH6	9185 MHz
CH7	9205
Ch8	9213 MHz
CH9	9221
Ch 10	9229
CH 11	9237
CH 12	9245

RAARR TESTER: AN/VPM-2 (10548182)

FREQ SELEct SWITCH POSITION	STALO OUT OUTPUT FREQUENCY
9205 mhZ	9145 MHz
9213 MHz	9153 MHz
9221 MHz	9161 MHz
9229 MHZ	9169 MHz
9237 MHZ	9177 MHz
9245 MHZ	9185 MHz

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE V. Output signal.

MICROWAVE POWER SOURCE 10556651 (ITEM 18) CHANNEL	. POWER AT DECOUPLED WIG TO TYPE H ADAPTER - FOR A +10 dBm OUTPUT SIGNAL AT END OF W201 (DBM)
1	
2	
3	
4	
5	
6 "	
7	
8	
9	
10	
11	
12	

1. Set the noise tester controls as follows:

   (1) Detune cavity (FREQ TUNE) control from normal operating range.

   (2) MODE switch to INF Or EXT TEST.

   (3) OPER/TEST switch to CAV TUNE.

j. The NOISE' INDICATOR meter should read within the limits defined by the black band. Adjust RF LEVEL control as rewired.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

k. Adjust FREQ TUNE control for signal peak, as indicated on NOISE INDICATOR meter. Adjust the RF level control setting as necessary to keep the meter reading within the scale limits during the peaking operation. Make final peaking adjustments with the fine tuning knob and lock the cavity. When a maximum signal peak has been obtained, adjust the RF LEVEL control for a maximum red indication, so that the meter is not pinned.

1. Repeat steps h. through k. for Channels 2 through 12 of the microwave power source (its 18, Table III).

me Set up equipment as shown in Figure 2 with microwave amplifier (item 32, Table 111) power switch OFF and power meter (items 6 and 8, Table III) connected to the W/G-to-Type N adapter (item 14, Table III)Via cable W201 and the 20 DB coaxial attenuator (item 34, Table 111). Set the power meter scale to +5 DBM. Ensure that the microwave power source (item 18, Table 1X1) is off by shutting down the +28 Vdc power supply (Item 1, Table 111). Set the variable waveguide attenuator (item 16, Table III) to maximum attenuation (60 DB).

CAUTION : Never apply power to the microwave amplifier Inmt unless its-titput Is terminated into a 50-ohm load. Damage to the traveling wave tube (TWT) could result.

n. Set the microwave amplifier (item 32, Table 111) power switch to allow thirty (30) minutes warm-up time before continuing test.

0. Set the microwave power source (item 18, Table 111) channel selector to Channel 1. Turn the +28 Vdc power source (item 1, Table III) to ON.

P. Adjust the gain control on the microwave amplifier (Item 32, Table III) so that the CATHODE CURRENT meter reads In the RATED POWER region.

q. Adjust the variable waveguide attenuator (item 16, Table III) so that the power meter (Items 6 and 8, Table III) reads zero DBM. Place microwave amplifier (item 32, Table 111) in the STANDBY mode.

r. Disconnect the power meter (items 6 and 8, Table 111) and 20 DB coaxial attenuator (item 34, Table 111) and connect cable w201 (Item 41, Table III) to the RF IN port of the noise tester.

s. Place the microwave amplifier (item 32, Table III) In the OPERATE mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

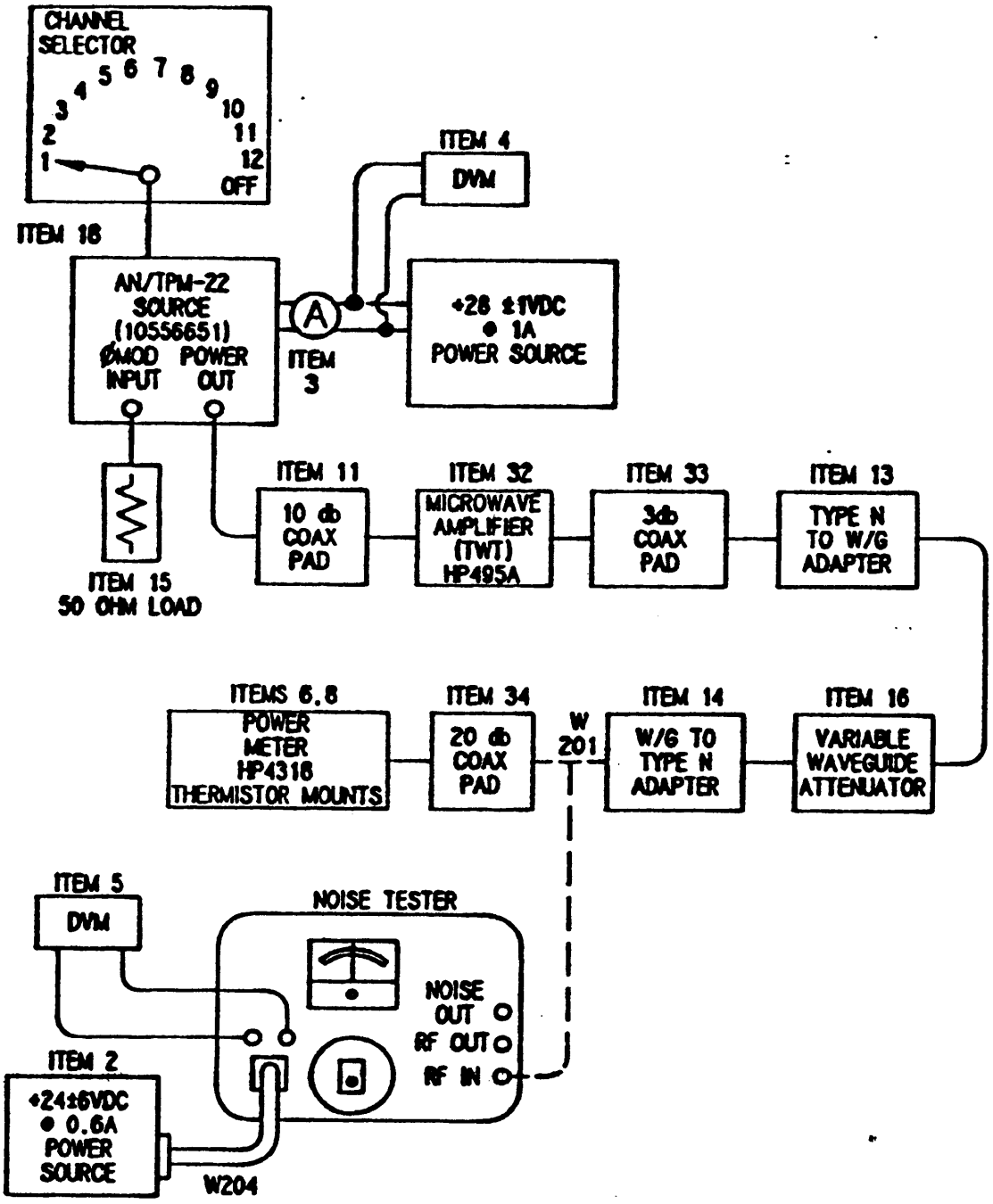


FIGURE 2. Test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

t. Repeat steps i. through k. for Channel 1 of the microwave power source (item 18, Table 111).

u. Repeat steps p. through t. for Channels 2 through 12 of the microwave power source (Item 18, Table III).

v* Set up equipment as shown in Figure 2, with microwave amplifier [item 32, Table XIX) power switch OFF, and the power meter (items 6 and 8, Table III) connected to the W/G-to-Type N adapter (item 14, Table III) via cable W201 and the 20 DB coaxial attenuator (item 34, Table S11). Set the power meter scale to +5 DBM. Ensure that the microwave power source (item 18, Table III) is OFF, by shutting down the +28 Vdc power supply (Item 1, Table 11X). Set the variable waveguide attenuator (Item 16, Table III) to maximum attenuation (60 DB).

CAUTION : Never apply power to the microwave amplifier input . unless Its output is terminated into a SO-ohm load. Damage to the traveling wave tube (TWT) could result.

w. Set the microwave amplifier (item 32, Table III) power switch to allow thirty (30) minutes warm-up the before continuing test.

x. Set the microwave power source (item 18, Table 111) channel selector to Channel 1. Turn the +28 Vdc power supply (item 1, Table 111) to ON.

Y* Adjust the GAIN control on the microwave amplifier (item 32, Table XII) so that the CATHODE CURRENT meter reads in the RATED POWER region.

20 Adjust the variable waveguide attenuator (item 16, Table III) so that the power meter (items 6 and 8, Table 111) reads +3 DBM. Place the microwave amplifier (item 32, Table III) in the STANDBY mode.

aa. Disconnect the power meter (items 6 and 8, Table III) and 20 DB coaxial attenuator (item 34, Table 111) and connect the W/G-to-Type N adapter (item 14, Table 111) to the RF IN port of the noise tester through cable W201• Set RF LEVEL control fully clockwise.

bb . Place the microwave amplifier (item 32, Table III) In the OPERATE mode. Allow one minute soak. Place the amplifier in the STANDBY mode.

cc. Repeat steps x. through aa. for all twelve channels of the microwave power source (item 18, Table 111).

dd . DiSConnect the noise tester from the test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

ee. Set up equipment as shown in Figure 1 and repeat steps g. through 1.

4.7.3.4 RF OUT.

a. Set up equipment as shown in Figure 1, with the power meter (items 6 and 8, Table 111) connected to the W/G-to-Type N adapter (item 13, Table 111). Set the channel selector (item 18, Table 111) to Channel 1.

b. Using Table V as a guide, adjust the variable waveguide attenuator (Item 16, Table XII) for a +10 DBM input signal to the noise tester, using the power meter (items 6 and 8, Table 111) as an indicator.

c. Connect the pOwer meter (items 7 and 9, Table III) to the RF OUT port of the noise tester.

d. Set RF LEVEL control to fully clockwise position.

e. Read level on power meter (items 7 and 9, Table 111) at RF OUT port. Signal level shall be -9 ± 2 DBM.

f. Set RF LEVEL control to fUlly Counterclockwise position.

9" Read level on power meter (items 7 and 9, Table III) at RF OUT port. Signal level shall be at least 12 DB down from the level noted in step 4.7.3.4.5.

h. Repeat steps d. through g. for Channels 6, 7 and 12 of the microwave power source (item 18, Table 111).

i. Disconnect power meter (items 7 and 9, Table III) from RF OUT port and connect spectrum analyzer (Item 31, Table III) in its place, through cable W202.

j. Set the microwave power source (item 18, Table III] channel selector to Channel 1.

k. Using Table V as a guide, adjust the variable waveguide attenuator (Item 16, Table 111) for +10 DBM input signal to the noise tester, as indicated on the power meter.

1. Set noise tester controls as follows:

(1) MODE switch to INTL TEST.

(2) OPER/TEST switch to BAND 2 OPERATE.

(3) Adjust REF LEVEL control to verify that the reference oscillator FM sideband level is adjustable between 27 and 33 DB below the carrier as indicated on spectrum analyzer (item 31, Table III) .

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

m. Repeat steps l.(2) and l.(3) (with the OPER/TEST switch set to the band under test) for BANDS 1 and 3.

4.7. 3.5 Noise out.

a. Connect the NOISE OUT port of the noise tester to the oscilloscope (Item 35, Table III) Input terminated with 50 ohms, using a BNC tee fitting (item 36, Table III) and a 50 ohm load (item 15, Table III).

b. Set the MODE switch to EXT *TEST* position.

c. Set the OPER/TEST switch to +12 VB

d. AdjuSt the REF LEVEL control for 0.8 VOlt peak-tO-~ak as indicated on the oscilloscope (item 35, Table III).

e. Disconnect oscilloscope (item 35, Table III) from the NOISE OUT port and *connect* J1 of the calibrated divider network (item 39, Table III) *in its* place. Connect the oscilloscope (item 35, Table III) to J2 of the calibrated divider network. Note: Do not use 50 ohm termination and BNC tee fitting. Set oscilloscope VERT CAIN switch to 5mV/Div and Time/Div switch as required for at least one cycle/D$v.

f. Set the divider network bandswitch to proper position for band-under-test and the INPUT-OUTPUT switch to OUTPUT position. Set noise tester OPER/TEST switch to position A for the band-under-test as Indicated in the following chart. Using the oscilloscope as an indicator. adjust REF LEVEL control to obtain exactly six divisions.

9" Set OPER/TEST switch to position B for the band-under-test as indicated In the chart below. Set divider network INPUT-OUTPUT switch to INPUT position. Observe oscilloscope for proper indication - 5.35 to 6.75 divisions vertical deflection. This indicates that the noise tester internal attenuators are properly adjusted.

OPER TEST POSITION

BAND	(ATTEN OUT) POSITION A	(ATTEN IN) POSITION B
1	+12VA	BD1 Test
2	+12VB	BD2 Test
3	MTR off	BD3 Test

h. Repeat steps e. through f. for the remaining two bands.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:34Z
Check the source to verify that this is the current version before use.

i. DiCOnneCt calibrated divider network (item 39, Table 111) from noise OUT port and COnneCt frequency counter (item 37, Table III) in its place.

j. Set MODE switch to EXT TEST.

k. Set OPEr/tEST switch to MTR OFF.

1. Adjust REF LEVEL for proper sIgnal 1eved at counter. Read frequency. This is the reference oscillator frequency for Band 3, which iS 1 MHz ± 10 percent.

m. Set OPER/TEST switch to +12VB.

n. Read Counter for reference oscillator frequency for Band 2 (43 KHZ ± 0 percent).

0. Set OPER/TEST Switch to +12VA.

P" Read counterI for reference Oscillator frequency for Band 1 (2 KHz ± 10 percent).

q" DiSCOnnect frequency counter (item 37, Table 1X1) from Noise Out port.

4.7.3.6 RF tuning controls.

a. Refer to 4.7.3.3 for verification of tuneabiilty *over* frequency range.

b. Connect equIpment as shown in Figure 1.

c. Set microwave power source (item 18, Table *XII*) channel selector to Channel 1.

d. Using Table *V* as a guide, adjust the variabl e waveguide attenuator (item 16, Table III) for +10 DBM input Signal to the noise tester *RF* IN port, using power meter (Items 6 and 8, Table III) as an indicator.

e. Set MODE switch to INTL TEST.

f. Set OPER/TEST switch to CAV *TUNE* position.

g. Adjust FREQ TUNE to channel in use. Make final adjustments (fine tuning) with the NOISE INDICATOR meter as an indicator of the true peak. Adjust the RF LEVEL control as necessary to keep indication on scale during tuning. When final peak is attained, adjust for full scale red Indication (not pinned). Lock the cavity dial.

h. Set OPER/TEST switch to BAL position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

i. Adjust BAL Adj control for center meter reading, I.e., within the limits defined by the black band.

3" Set OPER/TEST switch to PHASE SHIFT position.

k. Adjust PHASE ADJ control for center meter reading, i.e., within the limits defined by the black band . Repeat steps *h.* through k.

Repeat entire test for Channels 6, 7 and 12 on the microwave power source (item 18, Table III). Disconnect W201 from RF IN port.

4*7.3.7 system gain self test

a. Set MODE switch to INTL test.

b. Set GAIN ADJ control to fully clockwise position.

c. Set OPER/TEST switch to BAND l/Test.

d. Press OSC TEST pushbutton and while depressed, adjust REE LEVEL control so that the meter shows center tidication.

e. Release the Osc TEST pushbutton . The meter reading shall remain within blue band region, as indication of correct gain for the band tested (Band 1).

f. Repeat entire test for the BAND 2 end BAND 3 positions of the OPER/TEST switch.

4.7.3.8 Intgrated noise measurements. All tests detailed herein depend on the integrity of the bandpass filter bandwidths which are tested and verified per their respective drawings (see 3.2.1 for drawing list).

a. Connect equipment as shown in Figure 3, with the exception of the O MOD INPUT port of the 10556651 (item 18, Table III) which should have its 50 ohm termination cap still in place. Set the 10556651 channel selector to Channel 1.

b. Using Table V as a guide, adjust the variable waveguide attenuator (iten 16, Table III) for +10 DBM Input signal to the RF IN port of the noise tester, using the power meter (item 6, Table III) as an indicator.

c. set the noise tester controls as follows:

(1) MODE switch to INTL TEST.

(2) Set the OPER/TEST switch to BAND 3 OPERATE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

d. Using the RF spectrum analyzer (item 31, Table III) connected to the RF Out port of the noise tester as an indicator, adjust the REF LEVEL control so that the reference oscillator FM sideband is 30 DB below the carrier.

e. On the noise tester, detune the FREQ TUNE coarse control at least one channel (8MHZ) away from the actual frequency being measured.

f. Set the noise tester controls as fellows:

(1) OPER/TEST switch to BAL position. Adjust BAL ADJ control for a center line meter indication.

(2) OPER/TEST switch to CAV TUNE position. Adjust FREQ TUNE control to the rough *init* ial setting for Channel 1 of the microwave power SOUrCe (item 18, Table III).

(3) Adjust the FREQ TUNE controls, first coarse then fine, for mazimum red deflection as indicated on the NOISE INDICATOR meter. Adjust the RF LEVEL control as necessary to keep the meter reading on scale during peaking procedureo set rf level control so that meter reads full red deflection not planed-after attaining peak. Lock the cavity. Note: This step is extremely important; the peak must be true and the cavity locked to prevent instability.

(4) Sat the OPER/TEST switch to the PHASE SHIFT position. Adjust the PHASE AIM control for a center line meter indication.

(5) Set the OPER/TEST switch to BAND 3 OPERATE, or for the band-under-test. Recheck sideband level set in d. Adjust the GAIN ADJ control for a center line meter indication.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

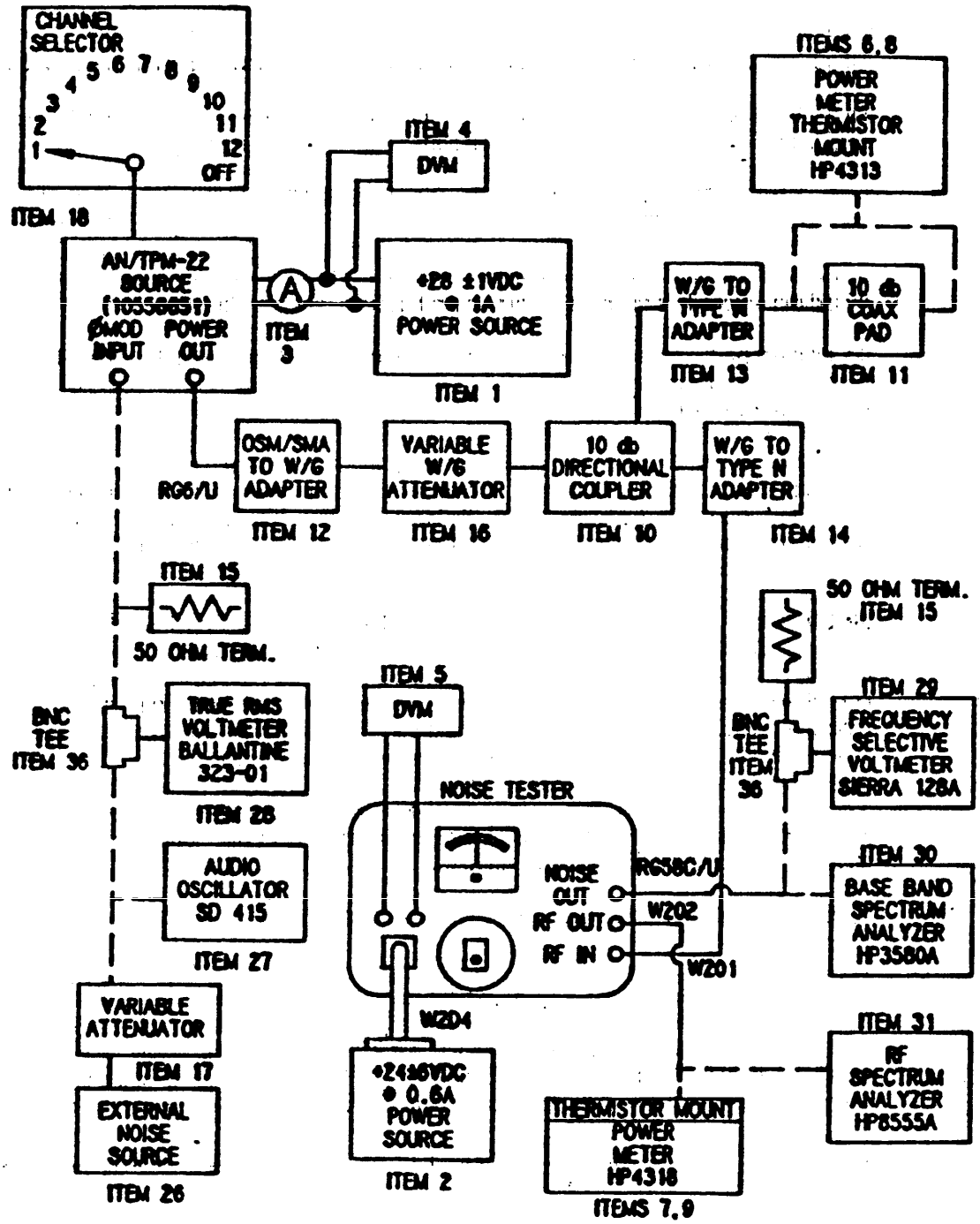


FIGURE 3. Integrated noise measurements test setup.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

g" Set MODE switch to INTL OPER. Allow 5 to 10 seconds for the NOISE INDICATOR *Meter to* stabilize. Observe the meter reading and grade source under test as follows: Red region Indicates a failed source, green or black region indicates a good source for the band tested.

h. Set the OPER/TEST switch to OPERATE/BAND 2, and the MODE switch to INTL TEST. Repeat d. through g.

i. Set the OPER/TEST switch to OPERATE/BAND 1, and the MODE switch to INTL TEST. Repeat d. through g. If the source under test grades good for all three bands, proceed with next step. Otherwise, substitute graded-good *source and* continue with test .

j. Disconnect the SO ohm termination (Item 15, "Table III) from the microwave power SUpply (item 18, Table III) 0 MOD INPUT port.

k. To the Q MOD INPUT port of the microwave power supply (Item lB, Table 111), connect the output of the external noise source (item 26, Table 11X) through a variable attenuator (item 17, Table XII) shown in *Figure* 3. Set the attenuator for maximum attenuation.

1. With the noise tester controls mat as indicated end using a graded-good source, the meter should be reading green, i.e., a passing Indication for the source of the band-under-test. Reduce the ● ttenuation of the external noise source (item 26, Table III) the amount necessary to provide a center line meter reading, i.e., a reading within the limits defined by the black band. Record the level indicated on the true MS voltmaster (item 28, Table 111) in the RMS black line noise space on data Sheet for band-under-test.

m. Set the *MODE* switch to EXT OPER.

n. Disconnect the external noise source (item 26, Table III) and its variable attenuator (item 27, Table III) and reconnect the audio oscillator (item 27. Table XII) in its place at the Q MOD INPUT port of the microwave power source (item 18 ,Table III. Do not disturb the level settings of the noise source or its attenuator.

0. Set the audio oscillator (item 27, Table 11X) for a 2 KHz ± 1 percent sine wave. InJect this calibration signal and observing the RF spectrum analyzer (item 31, Table III) at the RF OUT port, adjust the oscillator output level so that its FM sideband is 30 DB below the microwave power source (item 18, Table III) carrier.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:34Z
Check the source to verify that this is the current version before use.

p" Connect the base band spectrum analyzer (item 30, Table 111) to the NOISE OUT port. Observe the signal level (in DB below 1 volt) of the externally injected sine wave calibration signal. Enter this figure in the DBVC column (Column 3) of the Data Table for Band 1.

q. Disconnect the audio oscillator (item 27, Table III) from the Q MOD INPUT port of the microwave power source (Item 18, Table 111) and reconnect the external noise source (item 26, Table XIX) and its attenuator (item 17, Table III). Ensure that the level is the same as recorded in 1. on the true RMS voltmeter (item 28, Table III) .

r. For each frequency listed in Column 1 of the Data Sheet (Table VI), note and record the noise level indicated on the base band spectrum analyzer (item 30, Table 111). Enter result in DBVN Column 4A of Data Sheet. Clear analyzer screen, retrigger sweep and record new volume in COlumn 4B. Repeat this operation for all 5 raw data columns . Average these S values and enter final averaged DBVN value in Column 4F of Data sheet. A spread of more than 5 DB for one frequency over the five readtngs indicates a problem. Repeat *entire test* fox the frequency jn question. Repeat this step *for* all frequencies listd in column 1. Note: For Band 1, only one calibration signal (2 KHZ) is necessary to set a reference level, due to the fact that the cavity response for this band follows a predictable 6 DB/octave slope . This characteristic response holds true for Band 1 only, and is incorporated in the equation used for calculating Column 6 values for this band. These readings contain the data necessary for complete integrated noise power (DBCZ) measurement for the band tested.

s. Set nohe tester mode switch to INTL TEST and the 0~/TEST switch to CAV TUNE. Repeat f.(4) through f.(5). Set OPER/TEST switch to OPERATE BAND 2.

Using the RF spectrum analyzer item 31, Table III) as an Indicator, adjust the REF LEVEL control so that the reference oscillator EM sideband is 30 DB below the carrier.

Source: https://assist.dla.mil -- Downloaded: 2026t0-22T19:34Z
Check the source to verify that this is the currentversion before use.

TABLE VI. Noise tester integrated noise - Band 1.

S/N _____ DATE _____

CHAN _____ ENG._____

NOISE TESTER INTEGRATED
NOISE-BAND 1

SIDEBAND LEVEL (DBCR) = -30 dBc
RMS BLACK LINE NOISE = _____

1. FC, FN FREQ (Hz)	2. BW BANDWIDTH (Hz)	3. DBVC	MEASURED DBVN					AVG	5. DBF	6. DBCN (calculated)
			4A	4B	4C	4D	4E	4F		
360	30								-50	
400									-39	
500									-18	
600									0	
800										
1.000										
2.000										
2.500									0	
3.000									-20	
4.000	30								-50	



RELATIVE
METER READING

SPECIFICATION BAND 1 INTEGRATED NOISE = 54.3 dBc
MEASURED BAND 1 INTEGRATED NOISE = _____

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

u. Adjust the GAIN AM control for a center line meter reading.

v. Set MODE switch to INTL OPER with the external noise source (item 26, Table III) connected to the 0 MOD INPUT port of the microwave power source (item 18, Table XII). Adjust the attenuator (Item 17, Table III) to obtain a center line meter reading, i.e., a reading within the limits defined by the black band.

w. Set the MODE switch to EXT OPER.

x. Disconnect the external noise source (item 26, Table III and its variable attenuator (item 17,1 Table III) and connect the audio oscillator (item 27, Table 111) in Its place, at the 0 MOD INPUT port of the microwave power source (item 18, Table 111). Do not disturb the level settings of the noise ,source or its attenuator.

Y" Using Column 1 of Data Sheet (Table VII) as a frequency guide, inject a sine wave calibration signal and observing the RF spectrum analyzer (item 31, Table III) at the RF OUT port, adjust the oscillator output level so that its FM sideband is 36 DB below the microwave power source (item 18, Table III carrier.

-z.' with the base band spectrum analyzer (item 30, Table III) connected to the" NOISE OUT port, observe the Signal level (in DB below 1 volt) of the externally injected sine wave calibration signal. Enter this value in *the* DBVC side of Column 3 of the Data Table. Note: The base band spectrum analyzer has an u r bandwidth limit of 60 KHz. For higher frequency calbration signals, therefore, it is necessary to use a frequency slectlve voltmeter at the *NOTICE OUT* Port. Set freq sel Voltmeter INPUT control to 600 ohms. Perform the same operations as for the base band spe ctrum analyzer and enter results in DB in column 3 of the Data Table. Note: The freq sel voltmeter reading exhibits slight fluctuation - use average reading and enter value In Column 4F (Avg) of Data Sheet.

aa. Disconnect the audio oscillator (item 27, Table III) from the 0 MOD INPUT port of the microwave power source (item 18, Table III) and reconnect the external noise source (item 26. Table III) and its attenuator (it-17, Table 111). Ensure that the level is the same as recorded on the true RMS voltmeter (item 28, Table 111) in 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE VII. Noise tester integrated noise - Band 2.

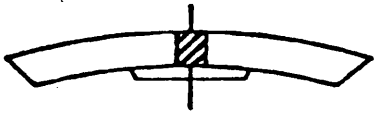
S/N _____ DATE _____

CHAN _____ ENG _____

NOISE TESTER INTEGRATED
NOISE-BAND 2

SIDEBAND LEVEL (DBCR) = -36 dBc
RMS BLACK LINE NOISE = _____

1. FC ,FN FREQ (Hz)	2. BW BANDWIDTH (Hz)	3. DBVC	MEASURED DBVN					AVG	5. DBG	6. DBCN (calculated)
			4A	4B	4C	4D	4E	4F		
1.500	100								-50	
2.000									-21	
2.500									0	
5.000										
10.000										
20.000										
43.000	100									
100.000	3.100									
200.000										
400.000										
600.000										
750.000									0	
900.000									-30	
1.000.000	3.100								-50	



RELATIVE
METER READING

SPECIFICATION BAND 2 INTEGRATED NOISE = 57.3 dBc
MEASURED BAND 2 INTEGRATED NOISE = _____

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

bb. On the base band spectrum analyzer (item 30, Table III) or frequency selective voltmeter (item 29, Table III) for frequencies above 50 KHz, observe the signal level of the external noise source (item 26, Table III) at the same frequency injected in y. Enter this value in Column 4A of the table.

cc. Set OPER/TEST switch to CAV TUNE and MODE switch to INTL TEST. Repeak cavity. Set OPER/TEST switch to BAL position, and adjust BAL ADJ control for center meter reading, if *required. Set* OPEr/TEST to phase shift position, and adjust PHASE ADJ control for center meter indication, if required. Reset MODE switch to EXT OPER and OPER/TESt switch to OPERATE BAND 2 or 3 as required. For base band analyzer, clear screen and retrigger sweep, and for frequency selective voltmeter, read average signal level of external noise source at the calibration signal frequency injected in y. Enter this value in COlumn 4B of Data Sheet. Repeat this entire step for the remaining three DBVN columns. Average these five values and enter result in COlumn 4EI of Data Sheet. Repeat s. through bb. for all frequencies listed In Column 1 of Data Sheet (Table VIII). These readings contain the data necessary for complete integrated noise power (DBCI) measurementt for the band tested.

dd. Set noise tester MODE switch to INTL TEST and the OPER/TEST switch to CAV TUNE. Repeat e. through f. Set OPER/TEST switch to OPERATE BAND 3. Repeat t. through dd.

ee. Using the data tabulated in COlumns 3 and 4 of each Data-Sheet, calculate COlumn 6 for each Table using the following three formulas where applicable.

BAND 1 EQUATION: Using base band spectrum analyzer (item 30, Table III), with FC = 2000 HZ, BW (bandwidth) = 30 HZ and reference sidebands = -30 DB:

1. DBW = DBVN - DBVC + 20 log FN - 50.79 + ADBF

Where: DBCN = Level of noise below the carrier at FC.
DBVC = Level of calibration signal below 1 volt.
DBVN = Level of noise at FC below 1 volt.
FN = Frequency of noise In Hz.
@BF = Filter attenuation characteristics (in DB) at Fc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

BAND 2 AND 3 EQUATION: For use with base band spectrum analyzer (item 30) and frequency selective voltmeter [item 29) where FN . = FC and reference sideboard level = .36 DB_0

2. DBCN = DBVN - DBVC - **10 log** BW - 36 + ΔDBF

where: DBCN = Level of noise below the carrier at Fc.
DBVN = Level of noise at FC below 1 volt.
DBVC = Level of calibration signal below 1 volt.
Bw = Bandwidth of measuring device in Hz.
DBF = Filter attenuation characteristic (in DB) at FC.

ff. Using the data computed for column 6 of each Data Table, calculate the measured Integrated noise power (in DB below the carrier) for each band-under-test (DBCX) using the following formula:

$$DBCI = 10\log \sum_{i=1}^{i=S-1} \left(\frac{10^{-1\ DBCNi}}{1-Mi} \right) FNi \left[\left(\frac{FN_{i+1}}{FNi} \right)^{(1-Mi)} - 1 \right]$$

where : DBCI = Integrated Noise Power (in DB) relative to the carrier in the measured band.

$DBCN_i$ = Level of tge Noise (in DB) in a 1 Hertz bandwidth relattve to the carrier at frequency FN_i

F N = Frequency away from the carrier where the noise was measured.

1 + 1 = Next frequency above F_{N_i} where the noise was measured.

s = Total number of measured frequencies.

‘ ¡ = $(DBCN_i - DBCN_{i+1}) / .(10 \log \frac{FN_{1+1}}{F\ N})$

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

TABLE VIII. Noise tester integrated noise - Band 3.

S/N _____ DATE _____

CHAN _____ ENG _____

NOISE TESTER INTEGRATED
NOISE-BAND 3

SIDEBAND LEVEL (DBCR) = -36 dBc
RMS BLACK LINE NOISE = _____

1. FC .FN FREQ (Hz)	2. BW BANDWIDTH (Hz)	3. DBVC	MEASURED DBVN					AVG	5. DBF	6. DBCN (calculated)
			4A	4B	4C	4D	4E	4F		
550	3.100								-50	
600									-36	
700									-12	
750									0	
1.000										
1.500										
2.000									0	
2.300									-25	
2.700	3.100								-50	



RELATIVE
METER READING

SPECIFICATION BAND 3 INTEGRATED NOISE = 56.8 dBc
MEASURED BAND 3 INTEGRATED NOISE = _____

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

gg. Compare the value of measured integrated noise for each band with the specification integrated noise value for that band as listed on each Data Table. The measured value shall be within ± 2 DB of the specification value. The specification values, as derived from MIL-S-48431 are as follows:

BAND 1 = -54.3 DBC
BAND2= -57.3 dBc.
BAND3= -56.8 dBc.

hh. Repeat the entire test for microwave power source (Item 18, Table 111) channels 6 and 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

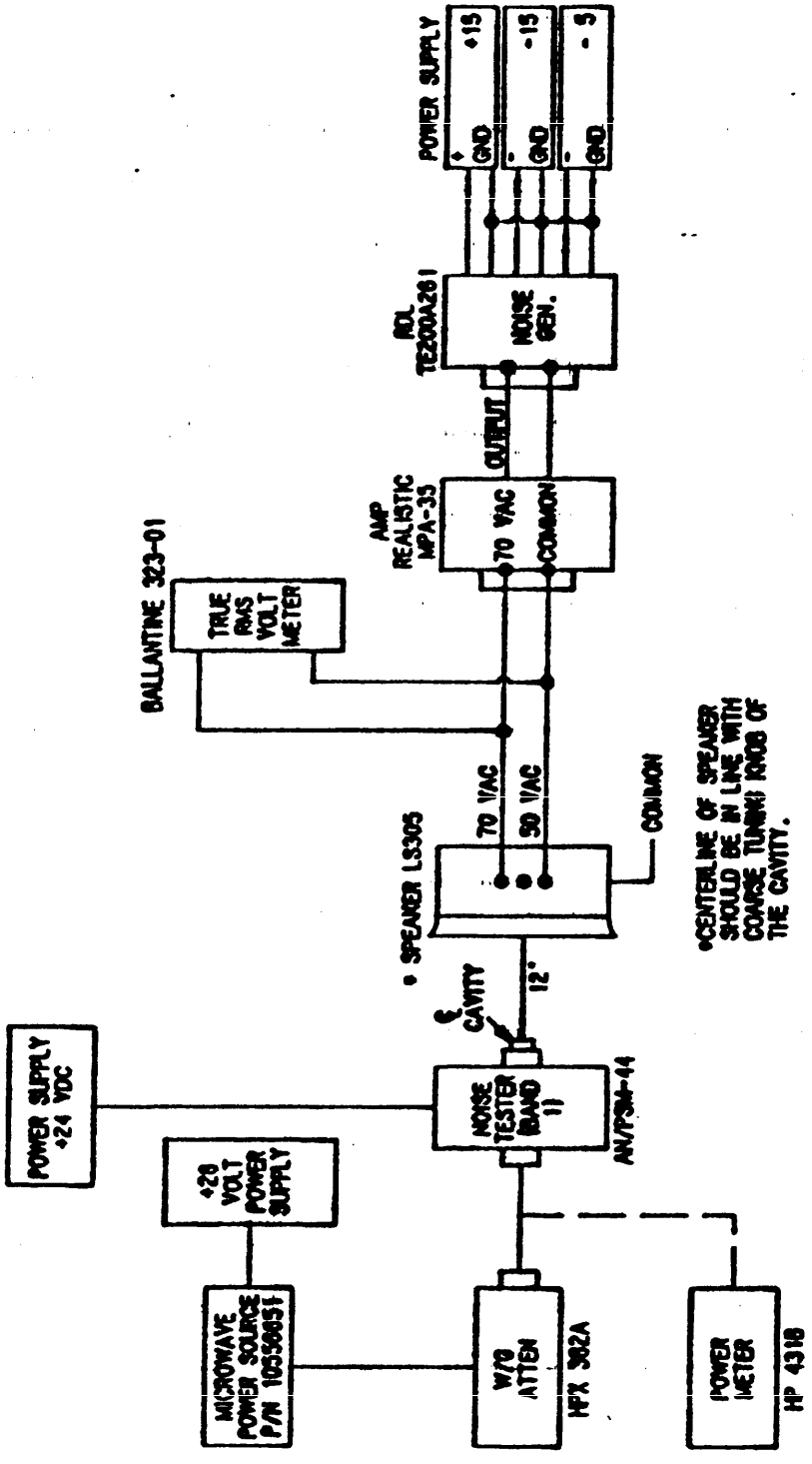


FIGURE 4. Acoustic test set-up "B".

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

4.7.3.9 Acoustic test.

a. Set up equipment as shown in Figure 4.

b. Apply +24 Vdc to noise tester. Allow 15 minutes warm-up.

c. Disconnect W/G attenuator from noise tester. Apply +28 Vdc to the microwave power source. Check for +10 DBM at Channel 6 using RF power meter. Reconnect W/G attenuator to noise tester.

d. Set the noise tester, MODE switch to INTL *TEST* OPER/TEST switch to CAV TUNE and RF LEVEL control 1/2 full scale.

e. Set EREQ TUNE at frequency being tested for maximum meter reading, using first COARSE then FINE tune.

f. Set RF LEVEL at RF LEVEL line on NOISE INDICATOR meter.

g" Detune FREQ TUNE at least four channels away from the actual charnel being tested.

h. Set OPEWTEST switch to BAL.

i. Set BAL ADJ to achieve center line reading on NOISE INDICATOR meter.

j. Set OPEWTEST switch to CAV TUNE.

k. Set FREQ TUNE at frequency being tested for maximum meter reading, using first COARSE then FINE tune.

1. Set RF LEVEL to RF LEVEL line on NOISE INDICATOR meter.

m. Set OPER/TEST switch to PHASE.

n. Adjust PHASE ADJ until meter is at center line.

0. Set OPER/TEST switch to BAND 1.

P" Depress and hold OSC TEST pushbutton and set RF LEVEL until meter is at center line. Release OSC TEST pushbutton.

q. Adjust GAIN ADJ until meter reads center line.

r. Set MODE switch to CNTL OPER.

s. NOISE INDICATOR meter should read GOOD.

t. Apply voltages to NOISE GENERATOR. Set Noise and Band 1 switches of-noise generator on. DiSConneCt true RMs voltmeter from LS30S speaker. Connect true RMS voltmeter to output terminal of noise generator. Adjust noise dial on noise generator for 60 ± 5 mV.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:34Z
Check the source to verify that this is the current version before use.

u. Disconnect true RMS voltmeter and connect noise generator output to MPA-35 amplifier aux. in port.

v. Set MPA-35 master volume to 10 and tone dial to 10 and adjust phono/aux dial for black center line reading on noise Indicator meter of AN/PSM-44. MeaSUre and record voltage on true RMS meter across the speaker. Spec limit 6.0 volts min.

4.7.3.10 Reference attenuator.

a. Connect equipment as shown in Figure 5.

b. Set noise tester RF-1evel fUlly counterclockwise and ON/OFF switch to ON. Allow 10 minutes warmup. Observe that power indicator illuminated.

co Set MODE switch to INTL TEST.

d. Set OPER/TEST switch to +12VA. The NOISE INDICATOR meter needle should remain in black band.

e. Set OPER/TEST switch to +12VB. The NOISE INDICATOR meter needle should remain in black band.

f. Set CAIN ADJ control fully clockwise.

g* Set OPER/TEST SWitCh to BD 1 TEST.

h. Depress and hold OSC/TEST pushbutton and set REF LEVEL control until the NOISE INDICATOR meter needle Is at center line. Release the OSC/TEST pushbutton.

i. The meter needle should remain within the blue band.

j.. Repeat steps h and i for the BD 2 TEST and BD 3 TEST positions of tie OPER/TEST switch.

k. Set MODE switch to EXT TEST and OPEWTEST switch to +12VA.

1. Set REF LEVEL control for a 1.0 V p-p signal as observed on oscilloscope ,

m. Set OPER/TEST switch to BD1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

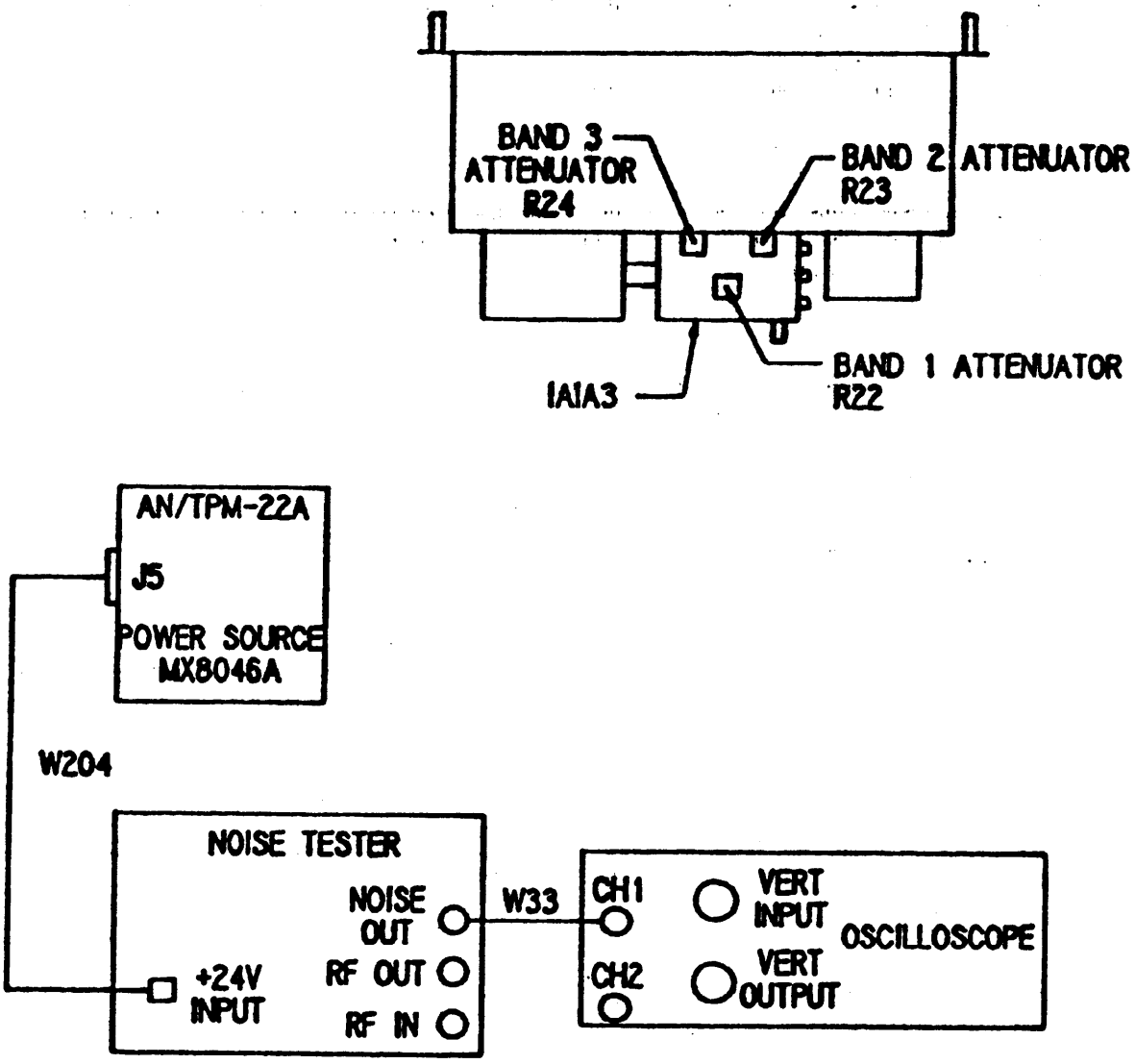


FIGURE 5. Test setup for checking reference attenuators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

n. Measure p-p signal and make the following computations for Band 1, Band 2 and Band 3:

P-P signal x 1.259 = Reference Attenuation (MAX)
P-P Signal x 0.794 = Reference Attenuation (MIN)

EXAMPLE:

Band 1 60 mV x 1.259 = 76 Reference Attenuation (MAX)
60 mV x 0.794 = 48 Referemce Attenuation (MXN)

Band 2 40 MV x 1.259 = SO Reference Attention (MAX)
40 mV x 0.794 = 32 Reference Attenuation (MIN)

Band 3 45 MV x 1.259 = 57 Reference Attenuation (MAX)
45 mv x 0.794 = 36 Reference AttenuatiOn (MIN)

o. Record the *results* of the above caputations, as per the followlng example, on the Reference Attenuation cards attached to the cable retaining cover:

Band 1 48 to 76 mV

Band 2 32 to 50 mV

Band 3 36 to 57 mV

4.8 Soldering. Soldering processes shall be Inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 to be specified in paragraph 3.8.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance~pecial Packaging Instruction PS11822700. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains Infoation of a general or explanatory natura that may be helpful, but is not mandatory.)

6.1 Intended use. The noise tester is designed to determine the integrated noise power of the Solid State X-Band Source (10548182/10556651) over the range of 600 Hz to 2 MKz, and grade it on a pass/fall, go/no go basis. The X-Band Source is a subassembly of the Receiver-Transmitter, RT-860/VPS-2 (a part of Radar Set AN/VPS-2) a subassembly of MX-8046A/TPM-22 (a part of Radar Test Set AN/TPM-22) and a subassembly of the R.F. Control Assembly of the Radar System Tester AN/VPM-Z.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date Of this specification.
b. Applicable stock number..
c. *First article* sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.
g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of Individual documents referenced (see 2.1.1).

6.3 Submission of contract inspection equipment designs for approval. Submit two copies of designs as required to: Conunander, U.S. ARmy AMCCOM, Attn: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawlngs. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also Include drawings prepared by, and identified as, U.S/\. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
FM sideband noise power
Radar Set AWVPS-2
Radar System Tester AN/VPM-2
Radar Test Set AN/TPM-22
Receiver-Transmitter, RT-860/VPS-2
vulcan *Air* Defense System (VADS)

6.6 Changes from Previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous Issue due to the extensiveness of the changes .

custodians:
Army-AR

Preparing activity:
Army-AR

(Project 1290 -A418)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
MIL-T-48621A (AR)	9 AUGUST 1991

3. DOCUMENT TITLE: TESTER, NOISE, X-BAND, AN/PSM-44: 11822700

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 — *Previous editions are obsolete.* — 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:34Z
Check the source to verify that this is the current version before use.